NATIONAL INSTITUTE OF TECHNOLOGY, PATNA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

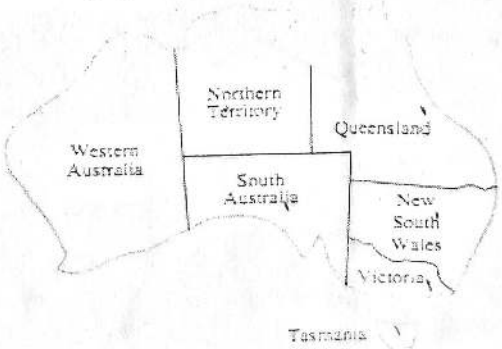
END-SEMESTER EXAMINATION – DECEMBER 2022

B. Tech. 5th Semester CSE

Date: 09th DECEMBER 2022(FN) *Time: 03 Hrs.* *CS5402 – ARTIFICIAL INTELLIGENCE*

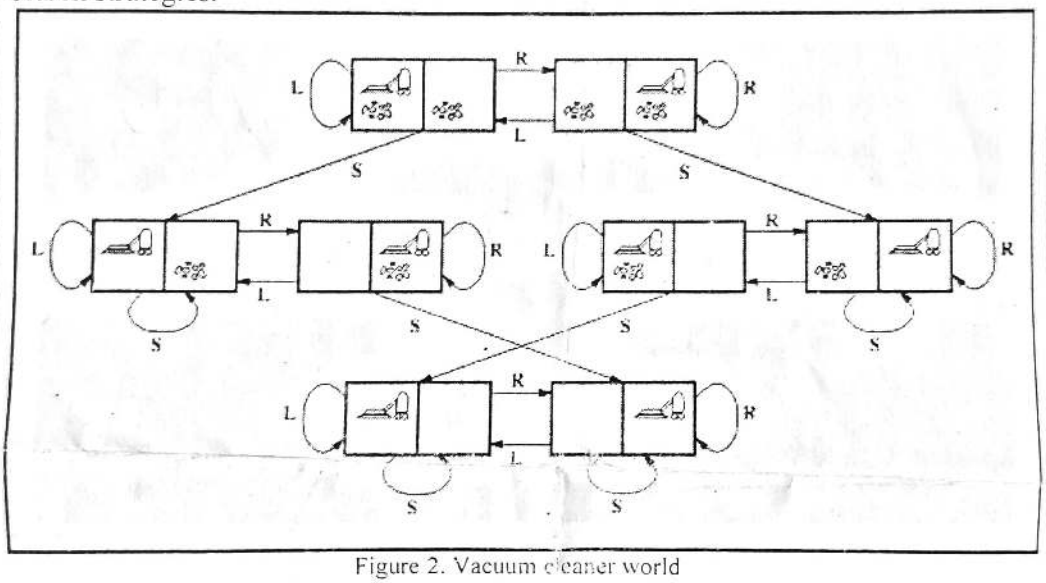
Max. Marks: 60

Instructions: *If any confusion, make your own assumption and mention it on your answer sheet respectively. The remaining instructions will be the same as per the Exam section of NIT Patna. All the questions are compulsory.*

Qus. no.	Questions	Marks	CO	BL
Q. 1.	Translate the following sentences into formulas in the First-order logic representation: i. One's husband is one's male spouse. ii. There is not any restaurant that sells popcorn, but cinemas do. iii. A sibling is another child of one's parents. iv. Male and female are disjoint categories. v. Parent and child are inverse relations. vi. All Romans were either loyal to Caesar or hated him. vii. Not all students like both AI and Compiler. viii. John likes all kinds of food. ix. Anything anyone eats and isn't killed by is food. x. John eats peanuts and is still alive.	[1.5*10 =15]	CO-1	I
Q. 2.	Write the pseudocode of the resolution algorithm of the propositional logic. What are the necessary steps used for converting the sentence into Conjunctive normal form (CNF) using a suitable example?	[05]	CO-4	V, VI
Q. 3.	Suppose that, we are looking at a map of Australia showing each of its states and territories as shown below in figure 1. We are given the task of coloring each region either red, green, or blue in such a way that no neighboring regions have the same color. Formulate this as a Constraint Satisfaction Problem (CSP) with the representation of constraint graph.  Figure 1. The principal states and territories of Australia	[05]	CO-1, CO-2	III, IV

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Q. 4.	Explain the Goal Stack Planning (GSP) method for the Block World problem with a suitable diagram and necessary functions. Also, explain partial order planning.	[06]	CO-1, CO-2	I, II, III
Q. 5.	Define the Bayes theorem and apply the Bayes' rule for examining the simple case and combining evidence with suitable examples.	[06]	CO-5	II, III
Q. 6.	Consider a modified version of the vacuum environment given below in figure 1, in which the agent is penalized one point for each movement. (ii). Comparing performance measuring components for all searching methods of the Uninformed search strategies.  Figure 2. Vacuum cleaner world a) Can a simple reflex agent be perfectly rational for this environment? Explain. b) What about a reflex agent with state? Design such an agent. c) How do your answers to **a** and **b** (in above) change if the agent's percepts give it the clean/dirty status of every square in the environment?	[07]	CO-1, CO-2	IV, V
Q. 7.	Differentiate between propositional vs. first-order inference logic rules. Draw the forward and backward chaining tree structure solutions (diagram) in first-order logic representation for the following knowledge-based automated reasoning agent and prove that West is a criminal: "The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American."	[08]	CO-3	II, IV
Q. 8.	Justify how the probability theory is a useful part of fulfilling the decision theory. Also, explain the six constraints that require any reasonable preference relation.	[08]	CO-5	V, VI

All the Best

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